

Convergence Tests



The ratio and comparison tests

- 1 Use the ratio test to determine whether $\sum_{n=1}^{\infty} \frac{n}{2^n}$ converges.
 - 2 Use the ratio test on $\sum_{n=1}^{\infty} \frac{n!}{n^n}$.
 - 3 Use the comparison test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n^2+3}$ converges, by comparing to $\sum \frac{1}{n^2}$.
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Alternating series and the p-series test

- 4 Determine whether each p -series converges or diverges:

a $\sum_{n=1}^{\infty} \frac{1}{n^3}$

b $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$
- 5 Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ converges, and classify it as absolutely convergent, conditionally convergent, or divergent.